Jaesung Choi

Korea Institute for Advanced Study (KIAS)
Bldg 8 Room 214, Hoegi-ro 85, Dongdaemun-gu, Seoul, 02455, Republic of Korea
✉ joseph9463@kias.re.kr

Employment

AI KIAS Junior Fellow Oct. 2025 – present
Korea Institute for Advanced Study (KIAS), Center for AI and Natural Sciences *Seoul, Korea*
Mentor: Prof. Jaesung Lee

Visiting Scholar Feb. 2026 – May 2026
Arizona State University, School of Electrical, Computer and Energy Engineering *Tempe, AZ, USA*
Host: Prof. Ying-Cheng Lai (Regents Professor)

AI Research Fellow Mar. 2023 – Sep. 2025
Korea Institute for Advanced Study (KIAS), Center for AI and Natural Sciences *Seoul, Korea*
Mentor: Prof. Jaesung Lee

Education

Ph.D. in Mathematical Sciences Aug. 2019 – Feb. 2023
Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Korea*

- Advisor: Pilwon Kim, Ph.D., Professor in UNIST
- Dissertation: *Oscillator-based reservoir computing and its applications*

M.S. in Mathematical Sciences Feb. 2017 – Aug. 2019
Ulsan National Institute of Science and Technology (UNIST) *Ulsan, Korea*

B.S. in Mathematical Sciences & Physics (double major) Mar. 2013 – Feb. 2017
Inha University *Incheon, Korea*

Research Interests

- **Machine Learning for Time Series:** Reservoir computing, autoencoder architectures, convolutional and attention mechanisms
- **Computational Methods:** Data augmentation and imputation algorithms, signal denoising, parameter extraction techniques
- **Nonlinear Dynamics:** Dynamical systems analysis, early warning indicators, critical transitions
- **Neuromorphic Hardware:** Physical reservoir computing implementations using Resistive Random Access Memory (RRAM) and Triboelectric Nanogenerators (TENG)

Publications

*(†equal contribution, *corresponding author)*

Journal Articles

2026

1. **J. Choi**†*, A.L. Chanu† and S. Awasthi†*. "Stochastic intracellular calcium dynamics show preserved structures identified by deep learning classification." *PLOS Computational Biology* 22.4 (2026): e1014240.
2. A.L. Chanu, Y. Park, **J. Choi**, Y. Cha, U. Lee, J. Moon and J. Park. "Human brain state classification via permutation entropy of EEG phase dynamics across consciousness levels and

inattentive-type ADHD." *Chaos, Solitons & Fractals* 209 (2026): 118416.

2025

1. **J. Choi**, and P. Kim. "Unsupervised reservoir computing for multivariate denoising of severely contaminated signals." *Knowledge-Based Systems* (2025): 114574.
2. **J. Choi**, and P. Kim. "Homotopy Reservoir Computing: Harnessing chaos for computation." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35.9 (2025): 093112.
3. **J. Choi**, and P. Kim. "Signal-Noise separation using unsupervised reservoir computing." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 35.8 (2025): 083136.
4. H. Park, D. Kim, I. Park, H. Nah, J. Yang, J. Yeo, **J. Choi**, J. Choi, H. Park and H. Choi. "Pd-modified microneedle array sensor integration with deep learning for predicting silica aerogel properties in real time." *ACS Applied Materials & Interfaces* (2025).

2023

5. **J. Choi**, E. Park, B. Jang and Y. Kim. "Reservoir concatenation and the spectrum distribution of concatenated reservoir state matrices." *AIP Advances* 13.1 (2023): 115110.

2022

6. **J. Choi**, and P. Kim. "Early warning for critical transitions using machine-based predictability." *AIMS Mathematics* 7.11 (2022): 20313-20327.

2020

7. **J. Choi**, and P. Kim. "Reservoir computing based on quenched chaos." *Chaos, Solitons & Fractals* 140 (2020): 110131.

2019

8. **J. Choi**, and P. Kim. "Critical neuromorphic computing based on explosive synchronization." *Chaos: An Interdisciplinary Journal of Nonlinear Science* 29.4 (2019): 043110.

Submitted Manuscripts

- J. Park†, **J. Choi**†, H. Kim, H. Park, J. Kim, J. Choi, S. Kim, M. Lee, and K. Heo. "Reconstructing hidden variables in chaotic and real-world system with a single-memristor based physical reservoir computing."
- J. Park, H. Kim, J. Choi, **J. Choi*** and M. Lee*. "A triboelectric nanogenerator-driven physical reservoir computing for intelligent cardiac anomaly detection"

Manuscripts in Preparation

- **J. Choi**, A.L. Chanu, and J. Park. "Self-supervised learning for distinguishing stochastic and chaotic dynamics."
- **J. Choi**. "Predictability-Calibrated SURE for self-supervised time-series denoising."

Presentations

Invited Talks

- "Reservoir computing and Echo state networks : Basics & Hands-on" *Tutorial* at 2026 UNIST Mathematical Imaging Lab Workshop, 2026.
- "Time series pattern discovery and harnessing complex systems for computation" Seminar at KIAS 2025 Winter Research Exchange Meeting
- "Reservoir computing: From chaotic system prediction to physical implementation." Seminar at Inha University, 2025.

- “Extracting predictable information from time series data using machine learning.” Seminar at National Institute for Mathematical Sciences (NIMS), 2025.
- “Extracting predictable information from time series data using machine learning.” 1st APCTP-SISSA Joint Workshop on AI and Theoretical Physics, 2024.
- “Multivariate signal-noise separation using unsupervised reservoir computing.” KSIAM 2024 Annual Meeting, 2024.
- “How reservoir computing differs from conventional neural networks and its potential applications.” Seminar at National Institute for Mathematical Sciences (NIMS), 2024.
- “Reservoir computing: Concepts and Implementations.” *Tutorial* at 2024 KPS Fall Meeting - Semiconductor Physics Department, 2024.
- “Strengths and recent research trends of reservoir computing.” Seminar at Korea University, 2024.
- “Harnessing dynamical systems for computation - The reservoir computing approach.” Seminar at Kyungpook National University, 2024.
- “Introduction to physical reservoir computing.” Seminar at Hanyang University (ERICA), Inha University, 2024.
- “Brief introduction to physical reservoir computing.” Non-Equilibrium Statmech Theory (NEST) Meeting at KIAS, 2024.
- Review on the paper “Reservoir computing with error correction: Long-term behaviors of Stochastic Dynamical Systems.” 11th Workshop on Nonequilibrium Fluctuation Theorems, 2023.
- “Constructing Efficient Reservoir Computers and Exploring Memory Capacity.”. Seminar at Korea Institute of Science and Technology, 2022.

Contributed Talks

- Review on the paper “Reservoir-computing based associative memory and itinerancy for complex dynamical attractors.” NEST Meeting at KIAS, 2024.
- “Exploring the computational potential of Titanium Dioxide Thin Films as physical reservoir computer.” KIAS Center for AI and Natural Sciences Workshop, 2024.
- “Denoising extremely noisy dynamical systems using unsupervised reservoir computing.” KSIAM 2024 Spring Conference, 2024.
- “Signal-Noise separation using unsupervised Echo state networks.” KIAS Center for AI and Natural Sciences Workshop, 2023.
- “Early Warning Indicators for noisy systems.” UNIST ML-based Early Warning Indicators Workshop, 2023.

Grants

KIAS Individual Grant (AP092902) Mar. 2025 – Feb. 2028

- Principal Investigator (PI), Center for AI and Natural Sciences, KIAS
- Research Subject: Predictive modeling research based on interaction between time series pattern discovery and neural network dynamical structures
- Amount: 13 million KRW per year

KIAS Individual Grant (AP092901) Mar. 2023 – Feb. 2025

- Principal Investigator (PI), Center for AI and Natural Sciences, KIAS
- Research Subject: Realization of reservoir computing framework based on complex systems
- Amount: 10 million KRW per year

UNIST Graduate Full Scholarship Mar. 2017 – Feb. 2023

Service and Outreach

Alternative Military Service

Technical Research Personnel Mar. 2023 – Feb. 2025

- Affiliation: Korea Institute for Advanced Study (KIAS)

Technical Research Personnel Mar. 2022 – Feb. 2023

- Affiliation: Ulsan National Institute of Science and Technology (UNIST)

Referee Services

Manuscripts reviewed for:

- Proceedings of the National Academy of Sciences (PNAS)
- Expert Systems With Applications
- Review of Scientific Instruments
- PLOS One
- Scientific Reports

Career Talks

- "Postdoctoral research experience in applied mathematics for undergraduate and graduate students." Math Night at UNIST, 2025.
- "Experiences as a researcher with a degree in mathematics." Seminar in Career Exploration Class at Kyung-Hee University, 2024.

Other Services

- Organizer and Session Chair, 2026 Winter Workshop on PDE and Applied Mathematics
- Session Chair, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2026
- Task Force Member, KIAS-Lite (Small Language Model) Development Project, Korea Institute for Advanced Study, 2025
- Session Chair, KIAS Lecture Series on Computational Neuroscience and AI, 2025
- Organizer, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2025
- Session Chair, Korea Institute for Advanced Study AI & Natural Sciences Workshop, 2024
- Organizer, UNIST Machine Learning-based Early Warning Indicator Workshop, 2023